WORKED SOLUTION

STEP II 1999, Question 3

Pure • Proof by Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q3

The key idea

Don't differentiate e^{-x^3} over and over from scratch. Instead, write the nth derivative as $e^{-x^3}S_n(x)$ and differentiate *that* once more. This gives a recurrence linking S_{n+1} to S_n and S_n', and every part of the question comes out of that one relation: the explicit polynomials, the induction, the degree and leading coefficient, and the final inequality.

Solution

The recurrence

By definition,

$$\frac{d^n}{dx^n}(e^{-x^3}) = e^{-x^3}S_n(x).$$

Differentiating once more with the product rule,

$$\frac{d^{n+1}}{dx^{n+1}}(e^{-x^3}) = e^{-x^3}S_n'(x) - 3x^2e^{-x^3}S_n(x),$$

and multiplying by e^{x^3} gives

$$\boxed{S_{n+1}(x) = S_n'(x) - 3x^2S_n(x).}$$

This holds for every $n \geq 0$, where $S_0(x) = 1$ (the zeroth derivative is the function itself).

S_2 and S_3

From $S_0 = 1$ the recurrence gives $S_1 = 0 - 3x^2 = -3x^2$, which you can also see directly. Then

$$S_2 = S_1' - 3x^2S_1 = -6x - 3x^2(-3x^2) = 9x^4 - 6x,$$

as required. Applying the recurrence again,

$$\begin{aligned} S_3 &= S_2' - 3x^2 S_2 = (36x^3 - 6) - 3x^2(9x^4 - 6x) \\ &= 36x^3 - 6 - 27x^6 + 18x^3, \end{aligned}$$

so

$$S_3(x) = -27x^6 + 54x^3 - 6.$$

The induction

We prove a stronger statement, because it costs nothing extra and answers the next question too:

$P(n)$: $S_n(x)$ is a polynomial of degree $2n$ whose leading coefficient is $(-3)^n$.

Base case. $S_1(x) = -3x^2$ has degree 2 and leading coefficient $-3 = (-3)^1$, so $P(1)$ holds. (So does $P(0)$, since $S_0 = 1$.)

Inductive step. Suppose $P(k)$ holds, so

$$S_k(x) = (-3)^k x^{2k} + (\text{terms of degree less than } 2k).$$

The derivative of a polynomial is a polynomial, and so is $3x^2 S_k$, so $S_{k+1} = S_k' - 3x^2 S_k$ is a polynomial. For its degree, look at the two pieces separately:

- S_k' has degree at most $2k - 1$;
- $-3x^2 S_k(x) = -3(-3)^k x^{2k+2} + (\text{lower terms}) = (-3)^{k+1} x^{2k+2} + \cdots$.

> **Ask yourself** Could S_k' cancel the top term of $-3x^2 S_k$?

No: S_k' has degree at most $2k - 1$, which is well below $2k + 2$, so it can't touch the x^{2k+2} term. Hence S_{k+1} has degree exactly $2k + 2 = 2(k + 1)$, with leading coefficient $(-3)^{k+1}$ (non-zero). That is $P(k + 1)$.

By induction, $P(n)$ holds for all $n \geq 1$. So $S_n(x)$ is a polynomial of order (degree) $2n$, and the coefficient of x^{2n} is $(-3)^n$. This agrees with S_2 ($9 = (-3)^2$, degree 4) and S_3 ($-27 = (-3)^3$, degree 6).

The final part

Suppose $S_n'(a) = 0$. Putting $x = a$ in the recurrence, the derivative term vanishes:

$$S_{n+1}(a) = S_n'(a) - 3a^2 S_n(a) = -3a^2 S_n(a).$$

Therefore

$$S_n(a)\, S_{n+1}(a) = -3a^2 \big(S_n(a)\big)^2 \leq 0,$$

since $a^2 \geq 0$ and $\big(S_n(a)\big)^2 \geq 0$. Equality happens exactly when $a = 0$ or $S_n(a) = 0$. That completes the proof.

Checks

- Leading terms: S_3 should start $(-3)^3x^6 = -27x^6$, and it does.
- Constant term of S_3: near $x = 0$, $\mathrm{e}^{-x^3} = 1 - x^3 + \cdots$, so its third derivative at 0 is $-3! = -6$. This matches $S_3(0) = -6$.
- A stationary point: $S_3'(x) = -162x^5 + 162x^2$ vanishes at $x = 1$. There $S_3(1) = 21$ and $S_4(1) = -3 \cdot 21 = -63$, so the product is $-1323 \leq 0$.
- We also checked $S_0, \ldots, S_9$ against the exact Taylor series of $\mathrm{e}^{-(a+h)^3}$ at thirty random rational points a, and tested the inequality at every stationary point of $S_1, \ldots, S_8$ in $[-4, 4]$.

Common slips

- Forgetting to say why the degree is *exactly* $2n + 2$. You need the leading coefficient to be non-zero and S_k' to be unable to cancel it, which is why it pays to carry the leading coefficient in the inductive hypothesis.
- Proving only "S_n is a polynomial" and then guessing the degree from a pattern. The question asks you to get the degree from the induction argument itself.

Beyond STEP

The polynomials here are cousins of the Hermite polynomials, defined by the same trick with e^{-x^2} (a "Rodrigues formula"). They satisfy recurrences like ours and turn up in probability and in the quantum harmonic oscillator. The final inequality is a first glimpse of how such recurrences control where the zeros and turning points of a whole family of polynomials lie. Boyce and DiPrima, *Elementary Differential Equations*, meets Hermite's equation in its chapter on series solutions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.